BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Exercise Sheet (5): Chapter 5 — Hydraulic Control Valves | Master Solution & Teaching Guide

Course: MEC 457 / MEC 431 — Fluid Power Systems & Control | **Examiner:** Dr. Ziad Ahmed Ibrahim
Teaching Assistant: Eng. Moh Zedan | **Mechatronics & Energy Departments** — Academic Year 2026/2027

---

1. Control of Power in Hydraulic Power Systems (Question 1)

Transmitted hydraulic power is the product of pressure and volumetric flow rate ($N_h = P \cdot Q$). Therefore, power control is achieved through two fundamental methodologies:

- **Throttling Control (Resistive Control):** Uses fixed-displacement pumps and inserts variable restrictions (control valves). Speed is controlled by dissipating excess pressure energy into heat. Simple and low-cost, but lower energy efficiency.
- **Volumetric Control (Displacement Control):** Uses variable-displacement pumps and/or motors. Flow rate and pressure are regulated directly by varying pump displacement (V_g) without throttling. Maximum efficiency with minimal heat generation, but higher initial cost.

2. Spool Valves vs. Poppet (Seat) Valves Comparison (Question 2)

Feature	Spool (Shearing) Valves	Poppet (Seated) Valves
Internal Leakage	Has radial clearance ($2 - 10 \; \mu\text{m}$) → continuous internal leakage ($Q_L \propto c^3$). Cannot hold loads without creeping.	Metal-to-metal line/conical contact → virtually **zero internal leakage**. Ideal for load locking.
Switching Response	Requires shifting over land overlap → slightly slower response time.	Very short travel (lift of a few millimeters) → extremely rapid opening and closing.
Contamination Sensitivity	High; fine silt particles can wedge into the radial clearance, causing spool sticking.	Low; self-cleaning poppet wipes the seat during closure.
Pressure Capability	Limited by sleeve deformation and sticking; up to 350 bar.	Can handle extreme pressures up to 700–1000 bar without jamming.
Multi-Port Routing	Can easily control 4, 5, or 6 ports simultaneously in a single body.	Typically 2-way or 3-way only; complex circuits require multiple poppet cartridges.

3. Function of Pressure Control Valves (Question 3)

Pressure control valves regulate system force and torque by throttling flow in response to pressure signals:

1. **Pressure Relief Valve:** Normally closed; limits maximum system pressure by discharging excess pump flow to tank.
2. **Pressure Reducing Valve:** Normally open; senses downstream pressure and throttles to maintain a reduced secondary branch pressure.

3. **Sequence Valve:** Normally closed; opens when primary circuit reaches setpoint to direct flow to a secondary actuator in a timed sequence.
4. **Counterbalance Valve:** Normally closed; maintains backpressure on vertical cylinder rod ends to prevent gravity loads from free-falling.
5. **Unloading Valve:** Automatically diverts low-pressure pump flow to tank at near zero pressure when high pressure is achieved by a hi-lo circuit.

4. Static Characteristics & Override Pressure in Relief Valves (Question 4)

4.1 Static Equilibrium Derivation

In a direct-operated relief valve, system pressure P acts on a poppet of area A_p opposed by an adjustable spring of stiffness k and initial compression x_0.

Cracking Pressure (P_{cr}): Pressure at which the poppet just begins to lift ($x = 0$): $$P_{cr} \cdot A_p = F_{spring,0} = k \cdot x_0 \implies \boxed{P_{cr} = \frac{k \cdot x_0}{A_p}}$$

Full-Flow (Opening) Pressure (P_o): When discharging maximum flow, the poppet lifts by displacement x: $$P_o \cdot A_p = k (x_0 + x) + F_{flow} \approx k (x_0 + x)$$ $$\boxed{P_o = \frac{k(x_0 + x)}{A_p} = P_{cr} + \frac{k \cdot x}{A_p}}$$

Override Pressure ($\Delta P_{override}$):

$$\boxed{\Delta P_{override} = P_o - P_{cr} = \frac{k \cdot x}{A_p}}$$

How to Reduce Override Pressure:
To minimize override pressure (make the static characteristic as flat as possible so $P_o \approx P_{cr}$):

1. **Increase Poppet Area (A_p):** Larger sensing area reduces the required pressure rise.
2. **Decrease Spring Stiffness (k):** Use a softer spring with a longer initial compression length (x_0).
3. **Use a Two-Stage (Pilot-Operated) Relief Valve:** The main spool spring handles only the tiny pressure drop across an orifice, giving a nearly flat characteristic over the full flow range!

5. Pilot-Operated Relief Valve Construction & Operation (Question 5)

A pilot-operated relief valve consists of two stages inside a single body:

- **Pilot Stage:** A tiny, direct-operated conical poppet with a light, adjustable high-pressure spring.
- **Main Stage:** A large balanced piston/spool held closed by a weak bias spring ($\Delta P_{bias} \approx 2 - 5 \text{ bar}$).
- **Communication Orifice:** Connects the inlet chamber to the spring chamber behind the main spool.

Operation: When inlet pressure is below setting, pressures on both sides of the main spool are equal through the orifice. The light spring holds the main spool closed. When pressure exceeds the pilot spring setting, the pilot poppet cracks, creating a small pilot flow through the orifice. The resulting pressure drop across the orifice creates an upward hydraulic unbalance that lifts the large main spool, dumping massive flow to tank with virtually zero override pressure!
Vent Port Function: By connecting the chamber above the main spool to tank via a small 2/2 solenoid valve, the main spool opens fully at just 2–5 bar, enabling easy electric pump unloading!

6 & 7. Pressure Reducers (Questions 6 & 7)

While relief valves control *inlet* pressure, a **Pressure Reducing Valve** controls *outlet* (downstream) pressure:

> **Working Principle:** The valve is **normally open**. Fluid flows freely from inlet port P to reduced port A. A internal pilot line senses downstream pressure at port A and directs it to the spool face opposed by an adjustable spring. If downstream pressure rises above setting, the spool shifts toward the closed position, throttling the passage to create a pressure drop that holds port A at the set pressure regardless of higher fluctuating inlet pressures!

8. Accumulator-Charging Valve (Unloader) (Question 8)

An accumulator-charging valve is a specialized pilot-operated unloading valve with an internal check valve and an area ratio differential piston. It automatically unloads the pump to tank at zero pressure when the accumulator reaches cut-out pressure (e.g., 200 bar). When accumulator pressure drops by approx. 15–20% due to machine usage, the valve recloses, directing pump flow back into the accumulator until fully recharged.

9 & 10. Directional & Pilot-Operated Check Valves (Questions 9 & 10)

- **Directional Control Valves (DCV):** Regulate the route taken by fluid streams, starting, stopping, reversing, or interlocking actuator motions.
- **Pilot-Operated Check Valve (POCV):** Functions as a normal leak-free check valve in the forward direction. In the reverse direction, reverse flow is normally blocked until an external pilot pressure signal is applied to an internal pilot piston, forcing the check poppet off its seat to allow reverse flow.
- **Primary Application:** *Zero-leakage load holding* in hydraulic cranes, lifts, and stabilizer outriggers. If a hydraulic hose bursts, the POCV stays mechanically shut, preventing fatal load drops!

11 & 12. Flow Control Valves (Questions 11 & 12)

- **Throttle-Check Valve:** Provides adjustable metered speed control in one direction, while the built-in parallel check valve allows unrestricted free flow in the opposite direction.
- **Series Pressure-Compensated Flow Control Valve:** Consists of a variable throttle in series with an automatic pressure-reducing compensator spool. The compensator continuously adjusts its throttling resistance such that the pressure drop across the metering orifice ($\Delta P_{orifice}$) remains strictly constant, making actuator speed totally independent of changing workload forces!

13 & 14. Flow Dividers (Questions 13 & 14)

- **Displacement-Type Flow Divider:** Consists of multiple identical gear or piston motor sections mechanically coupled to a single common shaft. Because all sections turn at the exact same rotational speed, input flow is divided into strictly equal outputs ($50\% / 50\%$), even under wildly unequal branch load pressures (efficiency up to 98%).
- **Spool-Type Flow Divider:** Uses two matched internal fixed orifices and a central floating spool. If pressure drops in one branch, the spool shifts dynamically to throttle the lower-load branch, maintaining equal flows.

Question 15: Comprehensive Circuit Calculation Problem

Given Data:
Pump: Displacement $V_g = 24 \text{ cm}^3/\text{rev} = 24 \times 10^{-6} \text{ m}^3/\text{rev}$, Speed $n = 1100 \text{ rpm} = 18.333 \text{ rps}$,
Volumetric efficiency $\eta_v = 0.94$, Mechanical efficiency $\eta_m = 0.95$, Hydraulic efficiency $\eta_h = 0.99$.
Directional Control Valve (DCV): Characteristic equation: $Q = 6 \times 10^{-7} \sqrt{\Delta P} \implies \Delta P = \left(\frac{Q}{6 \times 10^{-7}}\right)^2$.
Relief Valve: Cracking pressure $P_{cr} = 7 \text{ MPa} = 7 \times 10^6 \text{ Pa}$, Override pressure $= 0$.
Hydraulic Cylinder: Bore diameter $D = 50 \text{ mm} = 0.05 \text{ m}$, Rod diameter $d = 25 \text{ mm} = 0.025 \text{ m}$,
Extension velocity $v = 0.12 \text{ m/s}$, External resisting force $F = 10,000 \text{ N}$, Viscous friction factor $f = 2100 \text{ N}\cdot\text{s/m}$, No internal leakage.

Step-by-Step Mathematical Solution:

Step 1: Calculate Cylinder Areas and Flow Rates:
$$A_p = \frac{\pi}{4} D^2 = \frac{\pi}{4} (0.05)^2 = 1.9635 \times 10^{-3} \text{ m}^2 = \mathbf{19.635 \text{ cm}^2}$$ $$A_r = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}[(0.05)^2 - (0.025)^2] = 1.4726 \times 10^{-3} \text{ m}^2 = \mathbf{14.726 \text{ cm}^2}$$ Cap-end supply flow entering cylinder: $$Q_1 = A_p \cdot v = (1.9635 \times 10^{-3}) \times 0.12 = 2.3562 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{14.14 \text{ L/min}}$$ Rod-end return flow escaping cylinder: $$Q_2 = A_r \cdot v = (1.4726 \times 10^{-3}) \times 0.12 = 1.7671 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{10.60 \text{ L/min}}$$

Step 2: Calculate Pump Flow Rates:
Theoretical pump flow: $$Q_t = V_g \cdot n = (24 \times 10^{-6}) \times \left(\frac{1100}{60}\right) = 4.40 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{26.40 \text{ L/min}}$$ Actual (real) pump delivery: $$Q_p = Q_t \cdot \eta_v = (26.40) \times 0.94 = \mathbf{24.82 \text{ L/min}} = 4.136 \times 10^{-4} \text{ m}^3/\text{s}$$

Step 3: Calculate Pressure at Cylinder Rod End (P_2):
Flow Q_2 passes from the cylinder rod-end through the DCV back to tank ($P_{tank} = 0$): $$Q_2 = 6 \times 10^{-7} \sqrt{P_2 - 0} \implies P_2 = \left(\frac{Q_2}{6 \times 10^{-7}}\right)^2$$ $$P_2 = \left(\frac{1.7671 \times 10^{-4}}{6 \times 10^{-7}}\right)^2 = (294.52)^2 = 86,742 \text{ Pa} = \mathbf{0.867 \text{ bar}} = 0.0867 \text{ MPa}$$

Step 4: Dynamic Force Balance on Cylinder to Calculate P_1:
During steady extension: $$P_1 \cdot A_p = F + f \cdot v + P_2 \cdot A_r$$ $$P_1 \cdot A_p = 10,000 + (2100 \times 0.12) + (86,742 \times 1.4726 \times 10^{-3})$$ $$P_1 \cdot A_p = 10,000 + 252 + 127.74 = 10,379.74 \text{ N}$$ $$P_1 = \frac{10,379.74}{1.9635 \times 10^{-3}} = 5,286,346 \text{ Pa} = \mathbf{52.86 \text{ bar}} = 5.286 \text{ MPa}$$

Step 5: Calculate Pressure Upstream of DCV (P_3):
Flow Q_1 passes through the inlet port of the DCV into the cap end, experiencing a pressure drop: $$\Delta P_{DCV,in} = \left(\frac{Q_1}{6 \times 10^{-7}}\right)^2 = \left(\frac{2.3562 \times 10^{-4}}{6 \times 10^{-7}}\right)^2 = (392.70)^2 = 154,213 \text{ Pa} = 1.542 \text{ bar}$$ $$P_3 = P_1 + \Delta P_{DCV,in} = 5,286,346 + 154,213 = 5,440,559 \text{ Pa} = \mathbf{54.41 \text{ bar}} = 5.441 \text{ MPa}$$

Step 6: Determine Pump Pressure (P_p):
Notice that the actual pump flow is $Q_p = 24.82 \text{ L/min}$, while the cylinder demands only $Q_1 = 14.14 \text{ L/min}$. Because the FCV is in series, the excess flow ($Q_{relief} = Q_p - Q_1 = 10.68 \text{ L/min}$) must discharge continuously across the relief valve. With zero override pressure: $$\boxed{P_p = P_{cr} = 7.0 \text{ MPa} = \mathbf{70.0 \text{ bar}}}$$

Step 7: Power Losses in DCV and FCV:
Power Loss in Directional Control Valve: $$N_{DCV} = Q_1 (P_3 - P_1) + Q_2 (P_2 - 0)$$ $$N_{DCV} = (2.3562 \times

10^{-4}) \times (154,213) + (1.7671 \times 10^{-4}) \times (86,742) = 36.34 + 15.33 = \mathbf{51.67 \text{ W}}$$
Power Loss in Flow Control Valve (FCV): $$N_{FCV} = Q_1 (P_p - P_3) = (2.3562 \times 10^{-4}) \times (7.0 \times 10^6 - 5.441 \times 10^6)$$ $$N_{FCV} = (2.3562 \times 10^{-4}) \times (1,559,441) = \mathbf{367.4 \text{ W}}$$
Bonus Engineering Calculation — Power Wasted over Relief Valve: $$Q_{relief} = Q_p - Q_1 = 4.136 \times 10^{-4} - 2.356 \times 10^{-4} = 1.780 \times 10^{-4} \text{ m}^3/\text{s} = 10.68 \text{ L/min}$$ $$N_{relief} = Q_{relief} \cdot P_p = (1.780 \times 10^{-4}) \times (7.0 \times 10^6) = \mathbf{1246 \text{ W} = 1.25 \text{ kW}}$$

Summary of Question 15 Results:

Pressures: **P_1 = 52.86 bar (5.29 MPa)** | **P_2 = 0.867 bar (0.087 MPa)**

Upstream Pressure: **P_3 = 54.41 bar (5.44 MPa)** | **Pump Pressure: P_p = 70.0 bar (7.0 MPa)**

Flow Rates: **Q_1 = 14.14 L/min** | **Q_2 = 10.60 L/min** | **Pump Real Flow: Q_p = 24.82 L/min**

Power Losses: **N_DCV = 51.67 W** | **N_FCV = 367.4 W**